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Towards a framework for stimuli in interpreting and envisioning society in collaborative design futuring

Yuki Taoka[1]✉, Momoko Nakatani[1], Shiori Fujimaki[1] and Marianne Kinnula[2]

[1] Institute of Science Tokyo, Japan, [2] University of Oulu, Finland

✉ taoka.y.aa@m.titech.ac.jp

ABSTRACT: Collaborative design futuring has gained attention as an inclusive approach for envisioning future societies where external stimuli play a crucial role in stimulating imagination. Building on literature on design stimuli, psychological distance, and co-design, we propose and evaluate a five-layer framework through a multiple-case analysis of fifteen workshops and exhibitions. Through comparative analysis of different workshops, we explored how stimulus characteristics, such as modality, richness, and scenario, influence participant engagement and perceived psychological distance.

KEYWORDS: co-design, design tools, design creativity, collaborative design futuring, stimuli

1. Introduction

In an increasingly complex society, design futuring has been attracting growing attention. Practices and theories related to "designing for the future" vary widely (Sharma et al., 2024), encompassing approaches such as speculation (Light, 2021), foresight (Cuhls, 2020), and design futuring (Kozubaev et al., 2020). In this paper, we adopt design futuring as an umbrella term for these diverse approaches and position it as a means of facilitating the design of future exploration (Sharma et al., 2024). Recent studies have highlighted the importance of collaborative design futuring, in which participants collectively envision future possibilities (Farias et al., 2022; Kitazaki & Take, 2024; Ye & Duoduo, 2024). The development has led to a convergence of speculative design and co-design (Farias et al., 2022), the latter of which values the active involvement of non-designers in the design process (Sanders & Stappers, 2012). Non-designers' level of engagement in such processes can range from passive modes to more active contributions (Farias et al., 2022). Various methods for including non-designers emphasize the importance of stimuli (Groß & Mandir, 2024). In the process, a common strategy is to use external stimuli, prompts, or representations to spark imagination and broaden perspectives (Cuhls, 2020; Groß & Mandir, 2024).

Collaborative design futuring faces challenges in addressing long-term futures. As the temporal scope of the design target extends, non-designers often struggle with the perceived "remoteness" of the subject (Björgvinsson et al., 2012; Edeholt & Mainsah, 2014). The distant future, characterized by global issues such as climate change, is often perceived as unreal, which reduces individuals' motivation to act (Shabnam et al., 2021). While co-design has strengths in addressing local and concrete issues, it tends to face limitations when applied to more abstract or long-term futures (Björgvinsson et al., 2012; Edeholt & Mainsah, 2014). The "remoteness" is a key challenge to engaging non-designers in collaborative design futuring. The concept of "remoteness" is conceptualized in terms of psychological distance, which refers to the perceived gap between the self and a given object or context (Trope & Liberman, 2010). The closer this perceived distance, the more concretely people can engage with the topic (Trope & Liberman, 2010). Thus, psychological distance may affect how concretely participants can explore future scenarios, and external stimuli could play a significant role in controlling perceived psychological distance.

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The effect of stimuli has long been researched to examine the influence of external stimuli on designers' creativity, particularly in professional contexts (Blandino et al., 2023; Vasconcelos & Crilly, 2016). There is a clear lack of frameworks remaining for designing stimuli in collaborative design futuring settings involving participants from various backgrounds. This study aims to develop a stimulus selection framework for collaborative design futuring that supports the appropriate selection of stimuli based on the context of non-designers and the overall project context. We evaluate the framework's effectiveness through a set of empirical case studies. To guide the investigation, we set the following two research questions:

RQ1. What project-context factors should be considered when selecting stimuli in collaborative design futuring?

RQ2. What specific stimulus attributes influence participants' understanding of those stimuli and the resulting project-level and individual-level outcomes?

2. Related studies on design methods and stimuli in collaborative design futuring

To enable active engagement in collaborative design futuring, various design methods have been developed and applied (Kitazaki & Take, 2024). Workshops are commonly used methods in collaborative design futuring, which also serve as a core approach in co-design to enhance the quality of participation (Sanders & Stappers, 2012; Taoka et al., 2021). Within workshops, participants are guided through a movement across past, present, and future. Such temporal navigation is important in collaborative design futuring (Inayatullah, 2008). The influence of technology in shaping society plays a vital role in collaborative design futuring as the technologies become more complex and prominent (Østvold Ek et al., 2024). Its impact is discussed through stimuli, such as the use of artifacts or processes (Tsekleves et al., 2017), cards representing the application of cutting-edge technologies (Smith et al., 2024), and the systematic collection and presentation of weak signals of change (Saritas, 2013). The weak signal is fragments of information not yet mainstream but potentially meaningful for future scenarios (Groß & Mandir, 2024), which are classified through frameworks such as STEEP (Social, Technological, Environmental, Economic, and Political) and PEST (Politics, Economy, Society, Technology), which are commonly used to organize weak signals (Groß & Mandir, 2024). In workshops, card-based tools have been proposed for non-expert users to facilitate both technology comprehension and idea generation (Smith et al., 2024). In a review of card-based tools designed specifically for design futuring, cards served not only to explain specific technologies but also to help participants reconstruct meaning based on their own contexts (Ernstsen et al., 2023).

3. Method

3.1. Development of proposed framework

This study developed an analytical framework for stimulus design in collaborative design futuring. The framework was developed based on studies of co-design frameworks, creativity stimuli, and *psychological distance*. *Co-design* frameworks encompass a temporal sequence that encourages reflection on one's own values and aspirations through the relational framing of time (Sanders & Stappers, 2012). J.J. Lee et al. (2018) categorized four design choices in co-design activities derived from multidisciplinary collaboration projects: participants, project preconditions, project outcomes, and co-creation events. The participants' backgrounds are highlighted as significantly shaping diverse images of the future (Mahboob Kanafi et al., 2022). The participants' embodied accumulation of past social practices offers a valuable lens for understanding the differentiated responses of participants to the same design process (Vainionpää et al., 2024). *Creativity stimuli* aim to stimulate designers' idea generation, which is categorized as external or internal and is closely related to the design process (Crilly & Cardoso, 2017; Vasconcelos & Crilly, 2016). Key factors influencing external stimuli include modality of representation (e.g., text, image, video) and fidelity, while variables on the designer side include prior experience and disciplinary background (Vasconcelos & Crilly, 2016). *Psychological distance* refers to the degree to which an object or event feels distant from the self (Trope & Liberman, 2010). Psychological distance consists of four dimensions: temporal, spatial, social, and hypothetical (Trope & Liberman, 2010). Psychologically closer objects are perceived more concretely and vividly, whereas distant objects are perceived more abstractly (McDonald et al., 2015). It has been suggested that

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self-awareness and perceived similarity to oneself also play influential roles in psychological distance (Connors & Spangenberg, 2025). It is assumed that the closer the psychological distance, the more concretely future images can be envisioned, thereby increasing their quality.

The proposed framework categorizes factors that influence the design and selection of stimuli into three layers: *Aim, Constraint*, and *Design* (see Figure 1). The *Aim* layer refers to the goals the project seeks to achieve, which are divided into two types: *Project* and *Individual*, with different projects emphasizing one or both. *Project Outcome* refers to the tangible outputs or deliverables generated through the activity (Lee et al., 2018). *Individual Outcome* refers to the transformation, insights, or learning that participants personally experience (Akasaka et al., 2022; Lee et al., 2018). The *Individual Outcome* is critical when participants engage in design speculation based on intrinsic motivation.

The *Constraints* layer encompasses the practical limitations in implementing the *Aim*. These constraints include *Environmental* factors (e.g., in-person vs. online formats, location), *Time* limitations, and *Human Resources* (e.g., organizational structure and recruitable participant profile).

Both *Aim* and *Constraints* influence *Design*. The *Design* layer is further subdivided into *Program, Participants,* and *Stimuli*. Definitions of each component of the *Design* layer and hypotheses regarding their influence on psychological distance are summarized in Table 1. While *Stimuli* can be considered part of the *Program*, for clarity in this study, we treat them as separate components.

The *Program* and *Participants' Profile* directly influence the roles and representational forms of the *Stimuli*. The *Program* refers to the specific design elements of a project. Based on the related studies, we define the elements as *Theme* (Lee et al., 2018), *Target Year* (Trope & Liberman, 2010), *Activity Type* (*Lee et al., 2018*; *Sanders & Stappers, 2012*), and *Timeline* (Farias et al., 2022). *Participant* refers to characteristics of the expected participants. Based on discussions in design capital (Mahboob Kanafi et al., 2022) and prior stimuli research (Blandino et al., 2023; Vasconcelos & Crilly, 2016), the key elements are defined as *Literacy* (Mahboob Kanafi et al., 2022), *Historical Body* (Mahboob Kanafi et al., 2022; Vainionpää et al., 2024), *Design Experience* (Lee et al., 2018; Mahboob Kanafi et al., 2022), and *Motivation* (Akasaka et al., 2022). *Stimuli* are external information presented to participants, ranging from static forms, such as illustrations or text, to dynamic formats, such as video. *Stimuli* play a crucial role in triggering ideas, broadening perspectives, and influencing psychological distance. Table 1 outlines their classification and functions of stimuli. The categories are based on prior research of creativity stimuli (Blandino et al., 2023; Vasconcelos & Crilly, 2016) and impact of stimuli representation in collaborative design futuring (Yasuoka et al., 2024).

Psychological distance arises between participants and both the *Program* and the *Stimuli*. Program-induced distance depends on factors such as the topic and temporal scope of the future being addressed; typically, the more distant or unfamiliar the future scenario, the greater the psychological distance. In contrast, the distance induced by *Stimuli* varies based on the presentation format, content, and participant characteristics. Even when using the same *Program* or *Stimuli*, *Psychological distance* can differ depending on participants' level of understanding, interest, or changes in self-perception.

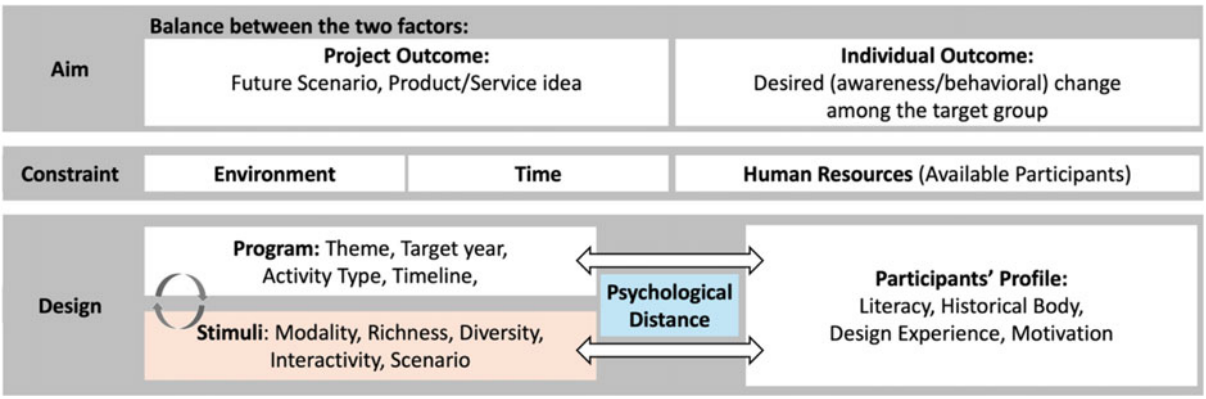


Figure 1. Factor Structure in designing collaborative design futuring programs

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Table 1. The factors in the design layer

Factor	Definition	Expected Impact on Psychological Distance
Program		
Theme	Focused theme or issue addressed in the workshop	Familiar issues tend to feel psychologically close, whereas topics like climate change feel more distant. Greater psychological distance requires more stimuli to facilitate understanding.
Target year	Temporal scope of the future envisioned in the activity	Psychological distance increases with temporal distance.
Timeline	Duration and frequency of workshops, and the specific activities and their time allocations within workshops	Longer and recurring sessions may reduce psychological distance. Covering all temporal dimensions, namely past, present, and future, may reduce psychological distance through reflections.
Activity Type	The degree to which participants can engage in the activity, including the mode of implementation	A higher level of engagement may reduce psychological distance.
Participants Profile		
Literacy	Participants' expertise and technical or societal knowledge	Individuals with higher topic literacy may understand the stimuli more easily and have lower psychological distance.
Design Experience	Experience with design activities or workshops	Individuals with greater design experience may be able to navigate and discuss ambiguous topics, even when psychological distance is high.
Historical body	The overall experiences, histories, and knowledge of people's lives	The more prior experience individuals have with the topic, the smaller the psychological distance they may experience.
Motives	Motivation for participating in the workshop	High intrinsic motivation may reduce psychological distance.
Stimuli		
Modality	The format in which stimuli are presented (e.g., text, image, video, real object).	The more personally relatable modality, such as a person speaking in a video rather than abstract statistical data, may reduce psychological distance.
Richness	The amount of information within stimulus representation. High richness offering detailed information.	More information may be preferable, as long as the amount does not become overwhelming.
Diversity	Thematic diversity of the stimuli. Greater diversity broadens participants' perspectives.	Greater diversity may reduce psychological distance across a broader range of individuals, as it is challenging to predict which element will resonate with each person.
Interactivity	Possibilities of interaction with the stimuli	Higher interactivity may reduce psychological distance.
Scenario	A narrative structure outlining a (hypothetical) sequence of events or actions under specific situational conditions	Including a scenario may reduce psychological distance.

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3.2. Evaluation

This study aims to exploratory evaluate the proposed framework by analyzing the contextual factors that influenced the selection of stimuli in past collaborative design futuring projects (RQ1). We also seek to clarify how the selected stimuli affected both project-level and individual-level outcomes (RQ2). Specifically, the analysis focuses on a series of initiatives conducted since 2022 within a university-based living lab currently being developed by the authors. This living lab serves as a platform for exploring new forms of co-creation within the university context, bringing together students, university researchers, companies, and government organizations to address social issues and envision desirable future societies. The effects of the stimuli were evaluated based on both participant interviews and observations during workshops. Multiple researchers analyzed each case, and the evaluations were synthesized collaboratively after individual analyses.

4. Result

4.1. Case studies applying the proposed framework

In each workshop conducted as part of this study, a variety of stimuli were designed and used to support participants, as shown in Table 2. Stimuli that conveyed social issues included cards, statistical data, and video-based scenarios. In the card format, social issues and user concerns were presented through combinations of text and illustrations, making them visually accessible and easy to understand. Statistical data included not only general macro-level indicators but also everyday figures directly linked to participants' lives, such as "daily vegetable consumption". Video scenarios incorporated themes such as alternative meats, veganism, and nutrition-focused food cultures to evoke imagination around future lifestyles. There were also attempts to use self-logged stimuli, such as participant-submitted photos and daily life logs, such as images of home gardening or records of refrigerator use. Interactive stimuli were also introduced. In particular, the Human Library format invited domain experts to serve as "living books," enabling participants to engage in dialogue and acquire knowledge through personal conversation (Yasuoka et al., 2024). As technical stimuli, VR and health-related technologies were offered for hands-on experience. Participants engaged in short interactions with the technologies before joining group discussions. In addition, technology-related stimuli were presented in card format, ranging from simple illustrations to more detailed cards with supplementary explanations. Some stimuli implicitly embed technical content into narrative formats. For instance, technologies related to alternative or lab-grown meat were integrated into storylines or four-panel comic strips, allowing participants to naturally associate the technology with broader social issues.

Table 3 summarizes a series of past workshops and exhibition-based projects using the proposed framework and an overview of the stimuli presented in each project. Due to space limitations, not all framework elements are included. Nonetheless, the table facilitates comparison across various Design Futuring activities, including exhibitions and workshops. In Projects 1–5, stimuli were designed to span multiple levels of psychological distance. In contrast, Projects 6 and onward, which were conducted under tighter time constraints, featured only a limited number of stimuli.

Table 2. Factors of stimuli used for the workshops and exhibitions

Name	Modality	Richness	Diversity	Interactivity	Scenario
Statistics	Graph and Text	Low	Adjustable	No	No
Exhibition	Artifacts	Low	Adjustable	No	No
Issue Card	Illustration and Text	Low-medium	Adjustable	No	No/Yes
Technology Card	Illustration and Text	Low-medium	Adjustable	No	No/Yes
Self-logging	Graphs, Pictures, etc.	Low-medium	Adjustable	Yes	No/Yes
Movie	Movie	Medium – High	Limited	No	Yes
Human Library	Human	Medium – High	Adjustable	Yes	Yes
4-panel comic	Illustration and Text	Low-medium	Adjustable	No	Yes
Experience	Experience	high	Limited	Yes	Yes

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Table 3. The overview of analyzed projects

#	Activity Type	Theme	Target year	AIM:	Participants' Profile	Representations
1-3	Workshops (3h x 4)	Food, child raising in a city, work style	2030	**Project:** Future scenarios **Individual:** Personalizing social issues and changing behavior	Companies, NPOs, local governments, University Students	Statistics, Video, Human Library
4	Workshops (3h x 4)	Sustainable life in the information society	2050	**Project:** Future scenarios and backcasting of research topics **Individual:** Becoming familiar with the backcasting method	Company (Engineering) and University Students	Statistics, Technology card
5	Workshop (2 h)	The aging society	2050	**Project:** Future scenarios **Individual:** Personalizing social issues and changing behavior	University Students	Video, Statistics
6-8	Workshop (0.5 h)	Toys and Healthcare	2050	**Project:** Future product prototypes **Individual:** Transfer of information between WSs	University festival participants (adults/ children)	Not used
9	Workshop (1 h)	Childcare	2035	**Project:** Future scenarios **Individual:** Getting people interested in the metaverse	Childcare workers	Experience
10	Workshop (12 hours)	Food	2040 (implicit)	**Project:** Animating future scenarios **Individual:** Personalizing social issues and changing behavior	Art Students/ Graduates	4-panel comic
11-12	Exhibition (0.25 h)	Food and Parenting	not specified	**Project:** Idea generation for future products and services **Individual:** Making social issues one's own	Companies, NPOs, local governments, University Students	Video, Artwork
13-14	Workshop (1 h)	Life in my city	in 2∼3 y.	**Project:** Thinking of ideas to solve problems in the future **Individual:** Acquire creativity confidence. Know how to think.	Teachers in junior high	Technology Card, Issue Card,Human Library
15	Workshop (1 h)	Promoting healthy aging	in 2∼3 y.	**Project:** Generate ideas for devices that will support future healthcare **Individual:** Increasing awareness of healthcare	Seniors	Experience

4.2. Framework application to illustrative projects

Figure 2 illustrates the application of the proposed framework to analyze actual decision-making in the design of two workshop programs (Projects 1 and 4, as listed in Table 3). Project 1's primary aim was to

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maximize *Individual Outcome*. Accordingly, the theme selected for discussion, which was the future of food, was intentionally set to be psychologically close to the participants. In contrast, potential future challenges, such as food loss and animal welfare, could be perceived with a greater psychological distance. One of the key *constraint*s was the voluntary nature of participation. The constraints required a workshop *design* that enabled participants to derive benefit and be motivated. Thus, in *designing* the workshops, we selected three types of stimuli, the Human Library, and self-logging, to reduce psychological distance with future-oriented topics. The statistical data were used to provide both a broad overview (e.g., the current state of climate change) and more relatable insights into daily life (e.g., average vegetable intake). The timeline of the session was also balanced, with roughly half of the time allocated to thinking about the future and the other half to reflecting on one's past and present.

Project 4 explored sustainable life scenarios in 2050, which *aimed* to prioritize the project outcome by specifically identifying future R&D topics for a corporate partner. Due to confidentiality concerns, the participant pool was restricted to company employees and students, thereby limiting diversity. However, because participants attended the workshop in their professional capacity, their engagement was high. Another major constraint imposed by the company was the fixed target year of 2050, which was often perceived as psychologically distant. Thus, we designed the workshop themes to have a broader thematic scope, allowing participants to select themes with a psychologically closer fit during the workshops. Given the longer temporal distance to 2050, statistical stimuli emphasized familiar, everyday data. Additionally, to help participants narrow the psychological gap, approximately 80% of the session was devoted to envisioning future scenarios, thereby encouraging a stronger sense of connection with future society.

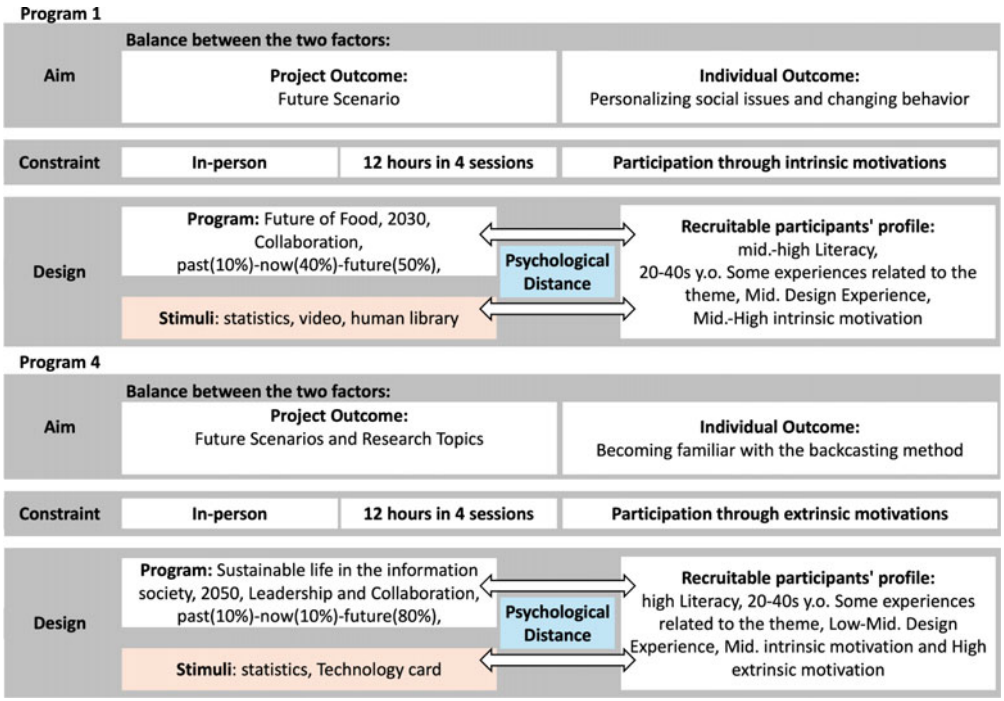


Figure 2. The analysis of two projects through the proposed framework

4.3. Observed effects of stimuli and psychological distance

Variations in participants' perceived psychological distance were observed even when the same content was conveyed through different forms or storytellers. For instance, in a video narrated by a child from Denmark, participants in Japan sometimes reported a strong impression despite the significant geographical distance (Program 1-3 in Table 3). Conversely, in cases where an adult from a different cultural background spoke a foreign language, such as Japanese scenarios viewed by participants in Hawai'i, both spatial and societal distances were perceived as large, and engagement tended to be lower. However, in some cases, these unexpected cultural differences evoked surprise and left a memorable impression (Program 4).

> *"I really resonated with the woman who was close to my age. Depending on the person, the point of resonance differed, and that really influenced the discussion." (quote in Program 1)*

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In the Human Library format, in which participants directly conversed with storytellers, participants gained a deeper understanding that extended beyond surface-level knowledge. Storytellers adjusted their narratives to participants' comprehension levels, resulting in more effective communication. Differences in participant responses across the storyteller's age and social role in the video and human library suggest that narrative structure and speaker selection are crucial considerations in stimulus design.

> *"It is different from just watching a video. Beyond the answers to our questions, I felt I got a broader sense of their life." (quote in Program 1)*

Familiar statistical data, namely those with low societal distance, such as average vegetable consumption, tended to evoke imagination and a sense of ownership in participants. In contrast, abstract and large-scale statistics (i.e., those with high psychological distance) were often less impactful and more challenging to integrate into discussions (Program 1-4).

> *"The large-scale stuff did not really register. [. . .] It was hard to imagine how forest loss and reduced crop yields would affect my life." (quote in Program 1)*

When data related directly to participants' own lives (i.e., zero psychological distance), such as photos of their gardens or refrigerator use, participants demonstrated strong comprehension and engaged in reflective reinterpretation of their daily practices. These stimuli also appeared to serve as prompts to reconsider their historical body.

In four-panel comic strips (Program 10), prior discussions were summarized visually. However, understanding of the content was limited without supplemental explanations; once the background context was provided, the participant's discussion became more active. In experiential stimuli (Programs 9 and 15), such as VR with a Head-Mounted Display, participants gained a tangible understanding of the technologies. In addition, participants were able to empathize with roles such as childcare workers, in which technology may be used in the future. The experience of technology appeared to promote idea generation, particularly when the workshop context was specific and concrete. In cases where technology was introduced via cards (Program 4), adult and university student groups actively used them as cognitive aids. However, among middle school students and teachers (Program 13-14), technical terminology and visual representations were difficult to interpret, necessitating facilitator supplementation in a Human Library format. Consequently, the impact of technology cards on the discussion was limited. Workshops with no or minimal stimuli (Program 6-8) tended to yield ideas confined to participants' existing frames of reference. The results suggest that such sessions may not sufficiently promote future-oriented thinking. In exhibition-based stimuli (Program 11-12), participants often perceived the installations as real technologies. However, many treated the displays as objects of appreciation rather than as media that reflected personally relevant experiences. Without mechanisms to encourage personal connection, these stimuli rarely triggered deep reflection or discussion. Still, when researchers actively engaged participants in conversation, the effect of the stimuli was amplified. Some participants spontaneously shared personal narratives rooted in their lived experiences, indicating that such stimuli can evoke memories and emotions that connect to individual life stories.

5. Discussion

This study explored how the presentation of diverse stimuli in Collaborative Design Futuring projects influenced participants' thinking and discussions. The proposed framework visually structures the parameters according to the aims and constraints of each project. In particular, psychological distance to themes and target year, aims, and time limitations were found to be critical in determining the quantity and complexity of feasible stimuli. This paper extended the co-design framework (Lee et al., 2018) to collaborative design futuring by introducing the concept of psychological distance. The findings suggest that stimuli can function as levers to modulate psychological distance. Stimuli may also play a role in shaping participants' self-perceptions of their historical bodies.

The analysis of stimuli used in the projects suggests that combining multiple stimuli with varying psychological distances contributed to enhanced project outcomes. As participants' understanding of the stimuli deepened, their psychological distance from the stimuli was reduced. Furthermore, this process led to increased comprehension of future societies and technological means, ultimately decreasing the overall psychological distance to the design subject. In addition, as evidenced by self-logging data, reflective engagement with stimuli enhanced self-understanding, leading to improved individual outcomes and a positive impact on the project's overall results. Differences in stimulus representation, particularly the attributes of narrators and degrees of interactivity in the video and human library, also influenced how

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participants perceived psychological distance and how broadly they could imagine alternative futures. These effects appeared to be mediated by participants' prior experience and accumulated knowledge, i.e., their historical body. In settings without external stimuli, participants' ideation tended to remain narrow, underscoring the importance of external stimuli in broadening everyday thinking, even when systematic methods such as horizon scanning (Cuhls, 2020) are not feasible. Furthermore, participants' literacy and prior workshop experience affected their ability to interpret and use the stimuli. This finding is consistent with previous research focused on professional designers, which has shown that participants' design experience (Blandino et al., 2023; Hu & Reid, 2018) and design process (Vasconcelos & Crilly, 2016) influence how stimuli are interpreted. In participatory design contexts, however, a wider range of participant attributes, such as contextual experience and technological literacy (Mahboob Kanafi et al., 2022), may also affect the impact of stimuli, underscoring the need for a more inclusive analytical framework.

Taken together, these findings suggest that when designing and selecting stimuli in collaborative design futuring, both project context and participants' profiles must be carefully considered. Stimuli should be evaluated not only for their effectiveness as communication tools, but also for their potential to modulate psychological distance. It should be noted, however, that this study is an exploratory investigation into the domains of technology and everyday life. The long-term effects of stimuli on participant learning and group discussion have not been thoroughly examined. Future research should empirically investigate how stimuli affect participants over time and how these effects propagate through the broader collaborative design process using a qualitative approach.

6. Conclusion

This study examined the role of stimuli not merely as tools for idea generation, but as means of supporting participants in interpreting current societal contexts and envisioning future society in collaborative design futuring. We proposed a three-layer framework for selecting stimuli and applied it across multiple case studies. The findings suggest that project aims and constraints shape workshop design, which includes program and stimulus selection. The preliminary evaluation indicated that psychological distance played a significant role in both the selection and the effectiveness of stimuli. Notably, psychological distance was observed to be relative rather than fixed, varying with the modality of the stimuli and the narrator's attributes. The participants' literacy levels also influenced their understanding and engagement with the stimuli. Furthermore, stimuli impacted not only project-level outcomes but also individual outcomes by fostering deeper self-understanding. While the findings are preliminary and limited to specific participant groups and thematic contexts, this study offers a multifaceted exploration of the importance and complexity of stimuli. It is intended as an initial step toward developing more systematic support tools for designing effective collaborative design futuring.

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